Initial Computational Analysis of Hydroxyl (OH) maser W3(OH)

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***Abstract*—**I analyze a public 506-second raw voltage (IQ) recording of the 1665 MHz ground-state hydroxyl (OH) line toward the massive star-forming region W3(OH), made in two linear polarizations with a single 6.2 m antenna of the Allen Telescope Array [1]. A Python pipeline is built in order to: average windowed FFT power spectra from the 3.9 GB-per-polarization SigMF files, remove the receiver bandpass with a running median, and convert frequency to velocity in the kinematic local standard of rest (LSRK). The averaged spectrum reveals OH maser emission between about -46.5 and -44 $\mathrm{km\,s^{-1}}$, with a main feature at $v_{\mathrm{LSR}} \approx -45.2$ $\mathrm{km\,s^{-1}}$ detected at roughly 15σ, a second feature at -46.2 $\mathrm{km\,s^{-1}}$, and weaker components near -44.9 and -44.1 $\mathrm{km\,s^{-1}}$. The main feature is stronger in the Y than the X polarization while the -46.2 $\mathrm{km\,s^{-1}}$ feature is not, which tentatively suggests partial linear polarization. With tracing W3(OH)'s motion as the main feature, a flat-rotation-curve kinematic distance of ≈ 3.8 kpc disagrees with the 1.95 kpc maser parallax distance [9], which resurfaces the well-known anomalous motion of this part of the Perseus Arm.

1 BACKGROUND

The hydroxyl radical (OH) has four ground-state hyperfine transitions near 18 cm: 1612, 1665, 1667 and 1720 MHz. In the dense gas around young massive stars, the 1665 and 1667 MHz "main lines" are often amplified by maser action. Instead of a broad, smooth line like the 21 cm hydrogen (HI) line, masers appear as a set of very bright, very narrow spectral spikes. Each spike traces a small, coherent region of gas moving at its own line-of-sight velocity. Because OH has a large Zeeman splitting, OH masers are also among the most direct probes of magnetic fields in star-forming regions.

W3(OH) is a very bright, compact HII region within the W3 complex, coincident

with OH masers [3] (Fig. 1). It lies in the Perseus spiral arm [9] and is one of the best-studied OH maser sources. Very Long Baseline Array (VLBA) observations have resolved hundreds of individual OH maser spots in it, including dozens of Zeeman pairs, with typical line widths of 0.15–0.38 $\mathrm{km\,s^{-1}}$ [2]. The source is also a benchmark for Galactic structure: its trigonometric parallax places it at 1.95 ± 0.04 kpc. That measurement resolved a long-standing factor-of-two disagreement with kinematic distances and showed that this part of the Perseus Arm has large non-circular ("anomalous") motions [9].

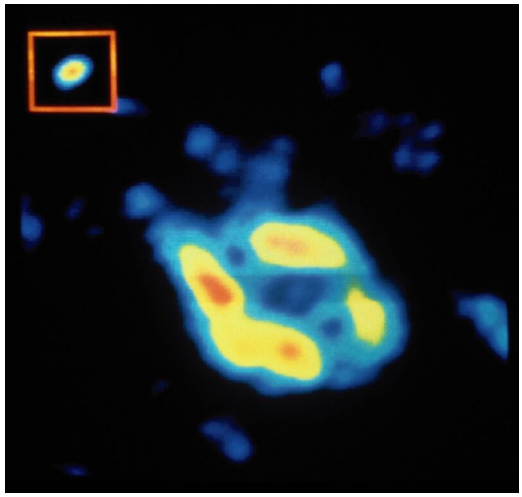


Figure 1—W3(OH) imaged with the Very Large Array in 2 cm (15 GHz) continuum on 1979 February 9 (investigators J. W. Dreher and W. J. Welch). Image: Legacy Astronomical Images, NRAO/AUI Archives [3].

The dataset analyzed here was recorded at the Allen Telescope Array (ATA), which has been fitted with USRP software-defined radios and a GNU Radio interface to make the array more accessible to people outside the radio astronomy community [7]. The recording was released publicly as raw IQ samples [1], not as finished spectra. Every processing step, from FFT to velocity, therefore has to be done by the user. That makes it a good self-contained exercise in radio spectroscopy, and a stepping stone toward processing my own small-dish observations made with low-cost SDR hardware.

1.1 Methodology

1.1.1 *Initial Data Analysis*

Dataset.—Table 1 summarizes the recording. The antenna's downconverter was tuned to 1665.4 MHz. Both linear polarizations were digitized by a USRP N321 locked to the observatory's 10 MHz reference and pulse-per-second signal. The two channels shared a local oscillator, and samples were written by a GNU Radio flowgraph and later converted to SigMF [1, 6]. Each polarization is a 3.9 GB file of interleaved 16-bit I/Q integers containing 971,942,220 complex samples (506.2 s). The 1.92 MHz bandwidth covers 1664.44–1666.36 MHz, so the 1667.359 MHz main line is *not* in the recording.

Table 1—Recording parameters [1].

Telescope	ATA, single 6.2 m antenna (1a)
Target (J2000)	W3(OH): $02^{h}27^{m}04.1^{s}$, $+61^{\circ}52'22''$
Line, rest frequency	OH $^{2}\Pi_{3/2}$, $J = 3/2$, $F = 1$–1; 1665.4018 MHz
Polarizations	Linear X and Y, separate files
Sample rate / bandwidth	1.92 Msps complex / 1.92 MHz
Format	SigMF, 16-bit integer IQ
Duration	506.2 s per polarization
Start time (from file name)	2022-06-21 19:36:55 (assumed UTC)

Visual inspection.—In `inspectrum`, each display column is a single FFT. At that sensitivity the band looks like pure noise, consistent with a maser that adds only a few percent to the system noise. A detection therefore requires averaging many FFTs.

Spectral averaging.—The pipeline memory-maps each data file and processes it in chunks. Each block of N_{FFT} complex samples is multiplied by a Hann window, Fourier transformed, and its power accumulated. With $N_{\mathrm{FFT}} = 16384$ the channel width is $1.92\,\mathrm{MHz}/16384 = 117$ Hz (0.021 $\mathrm{km\,s^{-1}}$ at 1665 MHz). The 506 s recording then yields about 59,000 averaged spectra per polarization. The pipeline also keeps a coarse dynamic spectrum by averaging 10 s blocks.

Bandpass removal.—The measured power in each channel is approximately

$$P(f) = G(f)\left[P_{\mathrm{sys}} + P_{\mathrm{line}}(f)\right], \tag{1}$$

where $G(f)$ is the unknown receiver gain and P_{sys} is the system noise. A 300 kHz running median of the averaged spectrum ignores the narrow maser features and estimates the baseline $B(f) \approx G(f)\,P_{sys}$. Before this fit, the outer 6% of each band edge (filter roll-off) and ± 1.5 kHz around the DC spike are excluded. Dividing by the baseline cancels the gain:

$$\frac{P(f)}{B(f)} - 1 = \frac{P_{line}(f)}{P_{sys}}, \tag{2}$$

so line strengths are expressed as a fraction of the system noise. Without an absolute calibration this ratio cannot be converted to kelvins or janskys.

Velocity axis.—Frequencies are converted to the radio-convention Doppler velocity $v = c(f_0 - f)/f_0$ with $f_0 = 1665.4018$ MHz. The topocentric velocity is then corrected to the LSRK frame. The correction combines Earth's rotation and orbital motion (barycentric correction computed with Astropy for the ATA site, the target position and the mid-recording time) and the standard 20 $\mathrm{km\,s^{-1}}$ solar motion toward $\alpha = 18^h$, $\delta = +30^\circ$ (B1900). For this observation the correction is $+14.9$ $\mathrm{km\,s^{-1}}$. It agrees with Astropy's independent `SpectralCoord` transformation to within 0.003 $\mathrm{km\,s^{-1}}$.

Expected sensitivity.—For a noise-limited measurement, the radiometer equation [8] predicts a fractional noise of $\sigma \approx 1/\sqrt{\Delta\nu\tau}$. For $\Delta\nu = 117$ Hz and $\tau = 506$ s this gives $\sigma \approx 0.0041$ per polarization, and ≈ 0.0029 for the average of X and Y.

Processing runs.—Table 2 lists the runs used in this report.

Table 2—Processing runs.

Run	Data	Time	Channel	Velocity frame
quick test	X	60 s	117 Hz	topocentric
`x_full`	X	506 s	469 Hz	topocentric
`x_fine`	X	506 s	117 Hz	LSRK
`xy_fine`	X + Y	506 s	117 Hz	LSRK

1.2 Interpretation

Detection.—The 60 s quick test already showed a single spike at about -60 $\mathrm{km\,s^{-1}}$ (topocentric), about 4σ above the noise. A matching bump appeared in the raw spectrum at $+335$ kHz from band center, the same frequency. Averaging

the full recording (`x_full`) raised the significance to about 12σ, as expected from the longer integration. In that run, however, the velocity window ended at -60 kms^{-1} while the peak sits at about -60.1 kms^{-1} topocentric. The reported peak value was therefore the edge of the line, not its maximum. This was a useful reminder that a velocity window must be set in the same reference frame as the axis.

Maser features.—Figure 2 shows the full-resolution, polarization-averaged spectrum. Table 3 lists the features. The emission spans about -46.5 to -44 kms^{-1}. The main feature at -45.2 kms^{-1} reaches ≈ 0.049 of the system noise. That is roughly 15σ, given an off-line noise of about ± 0.003, consistent with the radiometer prediction. A second feature appears at -46.2 kms^{-1}, a shoulder at -44.9 kms^{-1}, and a weak feature at -44.1 kms^{-1}. The -44.1 kms^{-1} feature appears independently in both polarizations, which supports it being real. The individual features are about 0.3–0.5 kms^{-1} wide. That is at or above the 0.15–0.38 kms^{-1} widths of individual W3(OH) maser spots measured at VLBI resolution [2], as expected: a single-dish beam blends many spots at similar velocities into each feature.

Table 3—Maser features in the 1665 MHz spectrum. Peak values are $P_{\mathrm{line}}/P_{\mathrm{sys}}$ (Eq. 2), read from the plots (uncertainty $\approx \pm 0.004$ per polarization).

Feature	v_{LSR} (kms^{-1})	X	Y	Mean
A	-46.2	≈ 0.025	≈ 0.024	≈ 0.024
B (main)	-45.2	≈ 0.043	≈ 0.055	≈ 0.049
C (shoulder)	-44.9	blended with B		
D	-44.1	faint	≈ 0.012	≈ 0.011

Polarization.—The two receiver channels differ in gain by about 2.8 dB (Fig. 2, panel 1), but Eq. 2 removes this. After normalization, feature A has the same strength in X and Y, while the main feature B is about 30% stronger in Y, consistently over several adjacent channels. A difference in system noise between the two channels would scale all features by the same factor. A mismatch on only one feature therefore suggests that feature B is partly linearly polarized. The difference is only about 2–3σ, so this is a tentative result.

Instrumental effects.—The X-polarization dynamic spectrum (Fig. 3, panel 3)

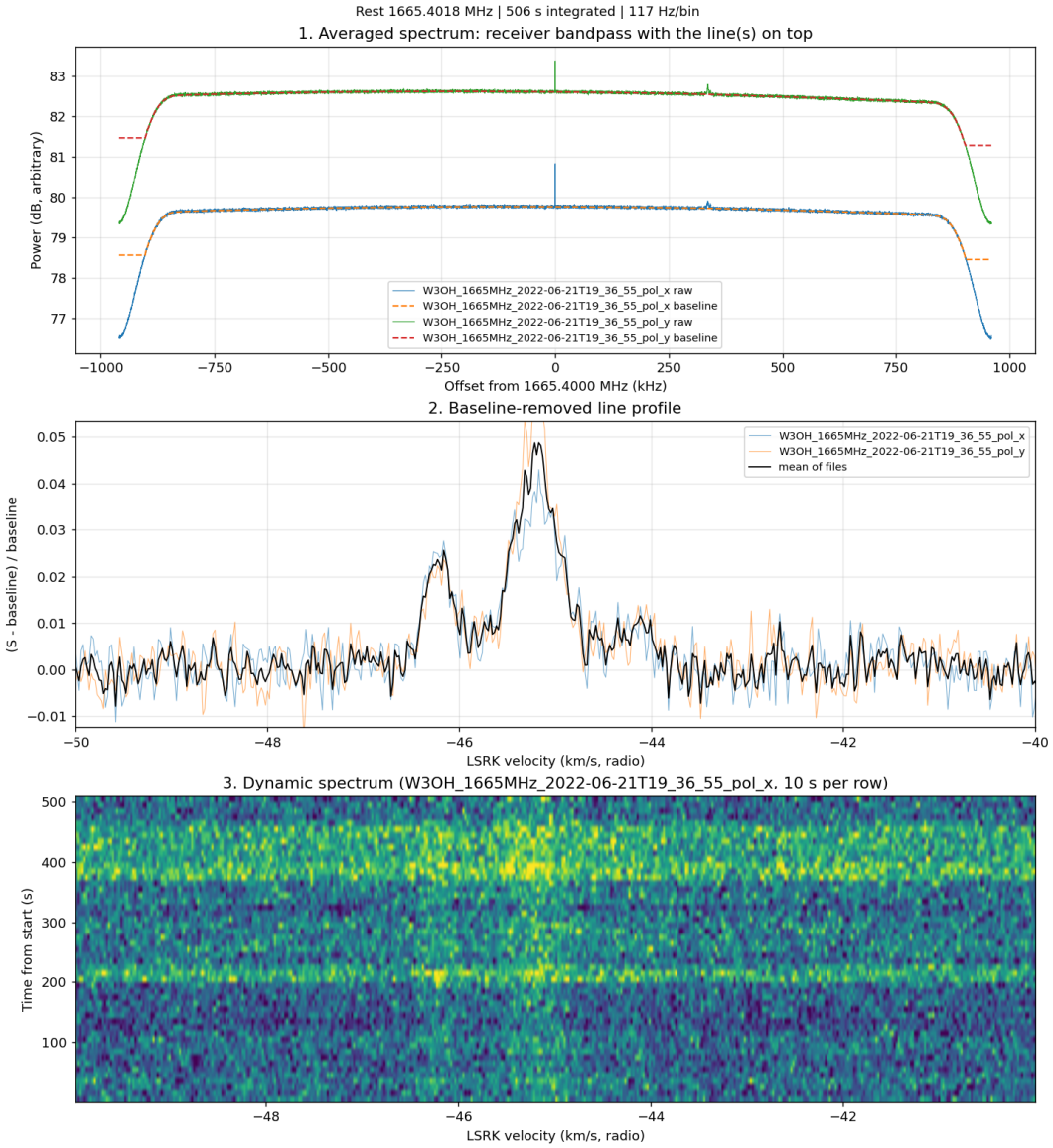


Figure 2—`xy_fine`: (1) raw averaged spectra of both polarizations, showing the receiver bandpass, the DC spike and the maser at +335 kHz; (2) baseline-removed profile versus LSRK velocity, with X, Y and their mean; (3) dynamic spectrum of the X polarization, 10 s per row.

shows bright horizontal bands at about 200–230 s and 370–470 s that brighten every velocity channel at once. These are not astronomical: they reflect changes in the X channel's total power, made visible because every row is divided by a single time-averaged baseline. The Y polarization's dynamic spectrum shows no such bands, and the maser appears there as a steady vertical stripe for the whole recording. The most likely cause is therefore a gain change in the X signal chain, although polarized interference cannot be ruled out.

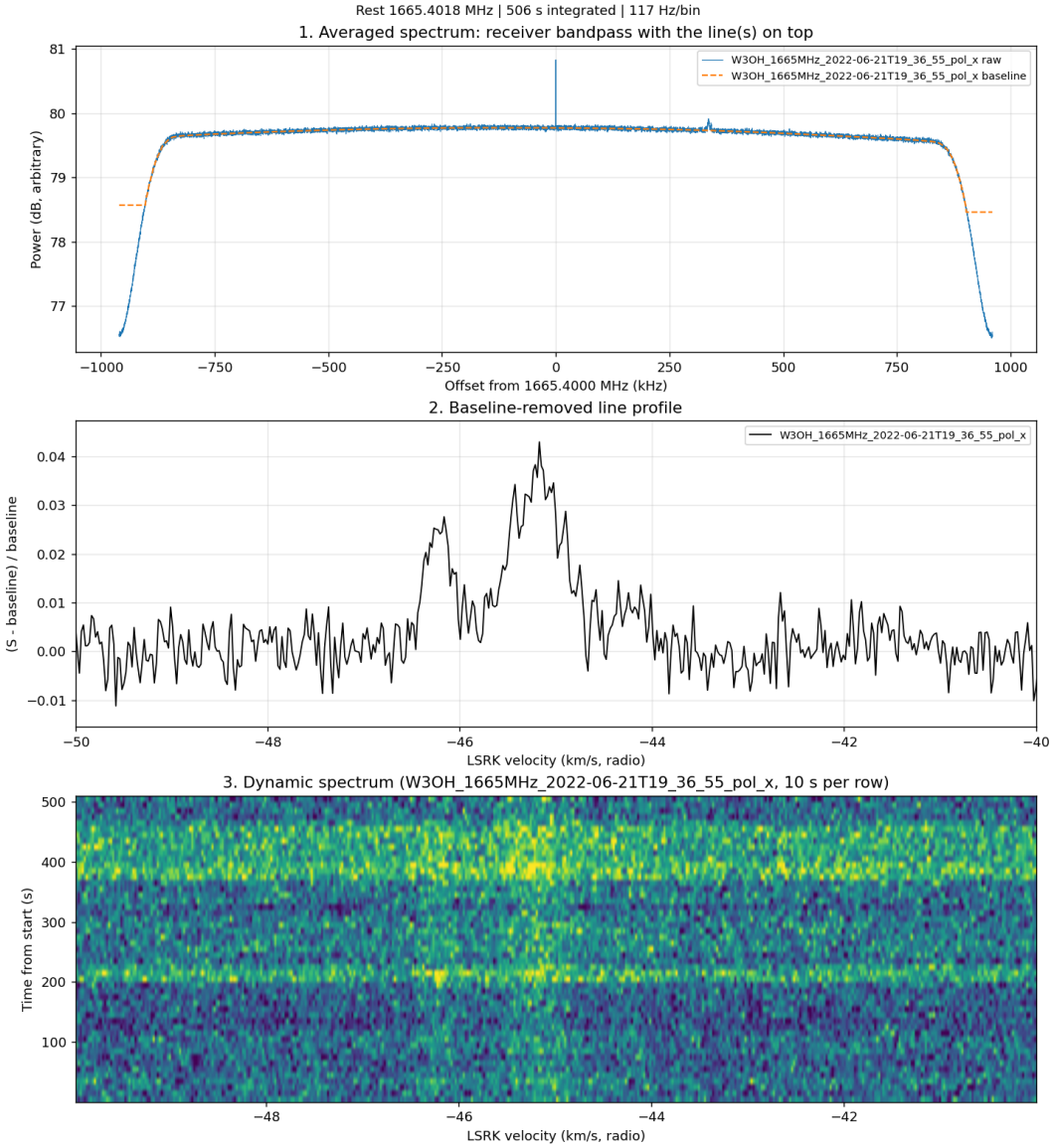


Figure 3—`x_fine`: X polarization only, 117 Hz channels, LSRK velocity. Panel 3 shows the broadband gain changes discussed.

Kinematic distance.—W3(OH) lies at Galactic coordinates $(l, b) = (133.95°, 1.06°)$. I assumed a flat rotation curve with $R_0 = 8.15$ kpc and $\Theta_0 = 236$ kms^{-1} [4]. Taking the main feature, $v_{LSR} = -45.2$ kms^{-1}, as representative of the source gives a Galactocentric radius of 11.1 kpc and a kinematic distance of ≈ 3.8 kpc. At the parallax distance of 1.95 kpc [9], pure circular rotation would predict only $v_{LSR} \approx -26$ kms^{-1}. The roughly 20 kms^{-1} difference is not an error in this measurement. It is the anomalous motion of this part of the Perseus Arm identified by Xu et al. [9], whose parallax showed that kinematic distances in this region are too large by about a factor of two. The exact size of the difference depends on the rotation curve and solar-motion parameters assumed, and on how well the maser velocities represent the systemic velocity of the region.

Limitations.—The measurements are relative (fraction of system noise), not flux-calibrated. A single dish provides no information about where the masers are on the sky. The 1667 MHz line, which is needed for main-line ratio diagnostics, is outside the recorded band. Polarization conclusions from X and Y amplitudes alone are limited: circular polarization, which is strong in OH masers, appears equally in both linear feeds.

1.3 Conclusion

Starting from raw IQ samples, I produced a calibrated-velocity spectrum of the 1665 MHz OH masers in W3(OH) from 506 s of single-dish ATA data. The emission is detected at about 15σ and resolves into at least three components between -46.5 and -44 kms^{-1} (LSRK), with the main feature at -45.2 kms^{-1}. The noise level is consistent with the radiometer equation, so the data appear to be noise-limited. The main feature's different strengths in X and Y hint at linear polarization. A kinematic distance computed from the maser velocity reproduces the known disagreement with W3(OH)'s parallax distance. This points to the famous Perseus Arm's anomalous motion. The exercise showed that public raw-voltage data, combined with a modest open-source pipeline, are enough to reach physically meaningful results. This holds as long as the instrumental effects (bandpass, DC spike, gain changes) and the choice of reference frame are handled carefully.

1.4 Future Direction

- **Full polarimetry and Zeeman splitting.** The X and Y channels shared a local oscillator, so cross-correlating them should allow full Stokes parameters to be

computed. Identifying oppositely circularly polarized Zeeman pairs would give the line-of-sight magnetic field directly. At 1665 MHz the splitting is about 0.59 kms^{-1} per milligauss, and Zeeman pairs are known in W3(OH) [2]. The uncalibrated phase offset between X and Y will limit the accuracy.

- **Quantitative line fitting.** Fit multiple Gaussian components to obtain velocities, widths and amplitudes with uncertainties, and compare the widths with the thermal width of OH (about 0.37–0.64 kms^{-1} for 50–150 K).
- **Robustness checks.** Measure the off-line noise formally against the radiometer prediction. Compare independent halves of the recording (a split-half test) to separate real features from instrumental ones.
- **Improved dynamic spectrum.** Normalize each time row by its own baseline to remove the X-channel gain bands, and display the X+Y average.
- **Variability.** Compare relative feature strengths with other epochs. For example, amateur observers at the Stockert telescope have reported changes in the relative intensity of W3(OH)'s two main 1665 MHz features [5].
- **Own observations.** Apply the same pipeline to my own SDR recordings of the 21 cm hydrogen line, and eventually attempt the W3(OH) 1665 MHz masers with a small dish.

2 REFERENCES

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